

Input Devices

Lesson 12 — Types, Functions and Nearest-Neighbour Traps

JKSSB Computer Awareness

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What Is an Input Device?

- ▶ An input device **feeds data and instructions** into a computer for processing.
- ▶ It converts a human action or a physical signal into **machine-readable** form.
- ▶ Common examples: keyboard, mouse, touchscreen, scanner, microphone, camera, barcode reader.
- ▶ Input is the first step of the input–process–output cycle.

Definition

An input device accepts data or control signals from the user or the environment and passes them to the CPU.

Classifying Input Devices

Category	Typical devices
Text input	Keyboard
Pointing input	Mouse, trackball, touchpad, joystick, light pen, stylus, touchscreen
Audio input	Microphone
Image / video input	Scanner, digital camera, webcam
Biometric input	Fingerprint, iris, face sensors
Code / tag input	Barcode, QR code, RFID

The same device can be classified by the type of data it captures.

Direct vs Indirect Input

Direct input

- ▶ The device captures the data itself
- ▶ Touchscreen, light pen, scanner, barcode reader, biometric sensor, microphone

Indirect input

- ▶ The user acts on the device to encode the data
- ▶ Keyboard, mouse, trackball, touchpad, joystick

Trap alert

A direct device captures data at source; an indirect device needs the user to enter or encode the data.

The Keyboard

- ▶ The keyboard is the most common **text input** device.
- ▶ The standard letter layout is called **QWERTY**.
- ▶ It has alphanumeric keys, function keys (F1–F12), modifier keys (Ctrl, Alt, Shift), and a numeric keypad.
- ▶ Each key press is converted into the corresponding character code.

Key idea

A keyboard turns a keystroke into a character code the computer can process.

Pointing Devices: The Mouse

- ▶ A hand-held pointing device moved on a flat surface.
- ▶ It controls an on-screen **pointer (cursor)** to point, click, double-click, and drag.
- ▶ It has buttons and usually a scroll wheel.
- ▶ An **optical/laser** mouse tracks movement with light; an older mechanical mouse used a rolling ball.
- ▶ Input device — PYQ-08.

Trackball and Trackpad

- ▶ **Trackball**: an upside-down mouse — a ball is rotated in place while the device stays still, saving desk space.
- ▶ **Trackpad / touchpad**: a flat, touch-sensitive pad used on laptops; slide a finger to move the pointer, tap to click.
- ▶ Both are **pointing input** devices.
- ▶ Do not confuse the trackball (rotating ball) with the mouse (device slides).

Touchscreen

- ▶ A display that is also an **input device**: touch it to select and enter data.
- ▶ The user touches the screen **directly**, so it is a direct input device.
- ▶ Common types are **resistive** and **capacitive**.
- ▶ Used in smartphones, tablets, ATMs, and information kiosks.

Why it is direct

The item touched on the display is the item selected — no separate pointer is needed.

Touchscreen vs Touchpad (Near-Neighbour)

Point	Touchscreen	Touchpad
Where you touch	On the display itself	A separate pad
Type of input	Direct	Indirect
Typical use	Phone, tablet, ATM	Laptop pointing

Trap alert

A touchpad is not a touchscreen. On a touchpad you touch a separate pad and the pointer moves on the screen.

Joystick and Light Pen

- ▶ **Joystick**: a lever that moves on two or three axes; used for games and some industrial control.
- ▶ **Light pen**: a pen-shaped device with a light sensor; pointed at a CRT screen to select or draw.
- ▶ Both are **pointing input** devices.
- ▶ A light pen is an input device, not an output device (TRAP-12).

Stylus and Digitizer Tablet

- ▶ **Stylus**: a pen-like tool used to write or draw on a tablet or screen.
- ▶ **Digitizer / graphics tablet**: converts pen strokes into digital coordinates.
- ▶ Used for freehand drawing, signatures, CAD, and handwriting recognition.

Keep the pair (TRAP-12)

A stylus and a digitizer tablet are **input** devices. A digitizer captures a drawing as data.

Scanners

- ▶ A **scanner** converts a paper document or picture into a digital image.
- ▶ Common types: **flatbed**, sheet-fed, handheld, and drum.
- ▶ Software such as OCR/OMR reads text or marks from the scanned image.
- ▶ A scanner is an input device (TRAP-12).

Scanned Codes: Barcode, QR Code, RFID

- ▶ **Barcode**: a 1D pattern of parallel lines read by a barcode scanner; used on product labels and ISBNs.
- ▶ **QR code**: a 2D matrix of black and white squares; holds more data; QR = **Quick Response**.
- ▶ **RFID**: Radio Frequency Identification; a tag and a reader exchange radio waves, with no line of sight needed; used in tolls, access cards, and inventory.

Trap alert

All three are **input** devices: they read a code or tag and feed the data into the computer.

Digital Camera and Webcam

- ▶ **Digital camera**: captures photographs as digital images stored on a memory card.
- ▶ **Webcam**: a small digital camera connected to a computer, used mainly for live video (video calls, streaming).
- ▶ Both are **image / video input** devices.
- ▶ A webcam is not used for printing; a printer is an output device.

Microphone (Audio Input)

- ▶ A **microphone** converts sound (voice or music) into an electrical signal that the computer digitises.
- ▶ Used for recording, voice calls, and speech recognition.
- ▶ Audio is input; a **speaker** produces audio output.

Pair to remember

Microphone = audio input; speaker = audio output.

Biometric Input

- ▶ **Biometrics** identifies a person from a physical or behavioural trait.
- ▶ Common traits: **fingerprint**, **iris**, **face**, voice, and palm.
- ▶ A sensor captures the trait and feeds it to the computer as input.
- ▶ Used for attendance, device unlock, and authentication.

Why it is input

The sensor reads a human trait and passes the measured data to the system.

Trap Alert: Input or Output? (TRAP-12)

- ▶ **Input:** light pen, punch-card reader, digitizer tablet, scanner.
- ▶ **Output:** plotter, drum printer, sound card.
- ▶ A monitor, printer, projector, and speaker are output devices.
- ▶ A keyboard, mouse, scanner, and microphone are input devices.

Trap alert

A plotter, a drum printer, and a sound card produce output. They are **not** input devices, however they are grouped in an option list.

Matching Special Devices (PYQ-09)

- ▶ **Haptic gloves**: worn on the hand to give touch/force feedback — output.
- ▶ **Tactile display**: presents touch sensations to the user — output.
- ▶ **Voice Response Unit (VRU)**: speaks responses to a caller — output.
- ▶ **Data glove**: worn to capture hand movement as data — **input**.

Trap alert

Haptic gloves, a tactile display, and a VRU give feedback to the user (output). A data glove captures hand movement (input).

Lesson 12: Recall

- ▶ An input device feeds data or control signals to the computer; input is the first step of the input–process–output cycle.
- ▶ By data type: text (keyboard), pointing (mouse, trackball, touchpad, joystick, light pen, stylus, touchscreen), audio (microphone), image/video (scanner, camera, webcam), biometric, and code/tag (barcode, QR, RFID).
- ▶ Direct input captures data at source (touchscreen, scanner, biometric); indirect input needs the user to encode it (keyboard, mouse).
- ▶ Near-neighbours: trackball (ball rotates) vs mouse (device slides); touchscreen (touch the display) vs touchpad (separate pad).
- ▶ TRAP-12: light pen, punch-card reader, digitizer, and scanner are **input**; plotter, drum printer, and sound card are **output**.
- ▶ PYQ-09: haptic gloves, tactile display, and VRU are output; a data glove is input.